Step 10: MFA Support for a Lost Phone

Board Activity 4 · Ticket IAM-0004 · Completed 5 November 2026

Dr. Sarah Chen (HVMC-1001, Doctors) lost the phone with her Microsoft Authenticator app. The help desk revoked her sessions and removed the lost phone's MFA method so she can register a new phone.

Prepared By	Sabitha S Karanavar
Date	November 5, 2026
Ticket	IAM-0004 — MFA support, lost device
Tool	Invoke-HvmcMfaSupport.ps1 (PowerShell 7, Microsoft Graph), actions Status / Reset / Verify
Performed by	helpdesk.admin (roles: Authentication Administrator + Reports Reader), not the Global Admin
Cost	$0 (Microsoft Entra ID Free)

Summary

Step	What	Status
0	Set-up: Sarah registered Microsoft Authenticator (her “old phone”)	✅ Done (4 Nov)
1	Status before the reset	✅ Done
2	Verify the caller: call-back to the phone number HR has on file	✅ Done
3	Dry run of the reset (nothing changed)	✅ Done
4	Reset: revoke all sessions, remove the lost phone's Authenticator	✅ Done
5	Sarah registers her new phone (3:52 PM)	✅ Done
6	Verify: 6 automatic checks — 6/6 passed (3:54 PM)	✅ Done
7	Audit log in the Entra portal	✅ Done

1. Getting the help-desk tool to sign in

The script needs the Microsoft Graph permission to manage authentication methods, which needs an interactive sign-in. Three problems came up before it worked. Each one taught something about identity security:

#	Problem	Cause	Fix
1	Script said *No account with employee ID HVMC-1001*	The browser sign-in had not finished; the script carried on without a connection and showed a misleading error	Script now stops with a clear message if it is not signed in or a permission is missing
2	*You don't have access to this*	The Global Admin is a personal Microsoft account (gmail). Without the tenant named, sign-in went to the wrong directory	Script signs in to the HVMC tenant by name (ID read at run time, never stored in the file)
3	Error **AADSTS530035**: access blocked by **security defaults**	Security defaults block the personal-account Global Admin from Microsoft Graph PowerShell	Kept security defaults ON. Created a dedicated help-desk account inside the tenant (next section)

2. Least privilege: a dedicated help-desk account

Instead of weakening security defaults or using the Global Admin for routine work, a help-desk account was created, the way a real hospital help desk is set up:

Item	Setting	Why
Account	HVMC Help Desk Admin (helpdesk.admin@<tenant>), IT, HVMC-ADM-01	Separate admin identity for help-desk work
Role 1	**Authentication Administrator**	Can reset MFA and revoke sessions for **normal users only**, never for admins
Role 2	**Reports Reader**	Can read the audit logs needed for the Verify step
MFA	Its own Microsoft Authenticator	Required by security defaults; every admin sign-in uses MFA
App approval	Microsoft Graph Command Line Tools approved for **helpdesk.admin only** (consent type: Principal)	Users cannot approve apps themselves (Step 7 hardening), so the Global Admin approved it for this one account
Initial password	Random, stored in a private file, changed at first sign-in, file deleted	Never shown on screen or in chat

> The Global Admin is now only used to set things up. The day-to-day MFA reset was done by an account that **cannot** change admins, roles or other settings, which is separation of duties and least privilege.

3. Step 1: status before the reset

Signed in as helpdesk.admin. Sarah's account is enabled and has one MFA method: Microsoft Authenticator on an iPhone 12 Pro (the phone reported lost). The audit history shows her registration on 4 November.

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Status
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-10-29 19:27 UTC (23):
    2026-10-29 20:39  Update PasswordProfile                    success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                               success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update PasswordProfile                    success  by Global Admin (IAM admin)
    2026-10-29 20:39  Reset user password                       success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update StsRefreshTokenValidFrom Timestamp success  by Global Admin (IAM admin)
    2026-11-02 19:13  Update user                               success  by Global Admin (IAM admin)
    2026-11-04 17:25  Update PasswordProfile                    success  by Microsoft password reset service
    2026-11-04 17:25  Update user                               success  by Microsoft password reset service
    2026-11-04 17:25  Update PasswordProfile                    success  by Microsoft password reset service
    2026-11-04 17:25  Reset user password                       success  by Microsoft password reset service
    2026-11-04 17:25  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:25  Reset password (by admin)                 success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)            failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                    success  by Microsoft password reset service
    2026-11-04 17:27  Change user password                      success  by Microsoft password reset service
    2026-11-04 17:27  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:27  Change password (self-service)            success  by sarah.chen@<tenant>
    2026-11-04 17:30  Update user                               success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:30  Update user                               success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                               success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                               success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                               success  by Azure Credential Configuration Endpoint Service
    2026-11-04 17:34  User registered security info             success  by sarah.chen@<tenant>
```

Figure 1. Status before the reset: 2 registered methods (Password + Microsoft Authenticator), MFA methods: 1.

4. Steps 2 and 3: verify the caller, then a dry run

Step 2 — identity check (most important). Before any reset, the help desk confirms the caller really is Sarah: *call-back to the phone number HR has on file*. An attacker who calls pretending to be a doctor who "lost their phone" is a common way to take over accounts (help-desk social engineering), so this check is required: the script refuses `-Apply` without `-VerifiedBy`.

Step 3 — dry run. The script shows exactly what it would do and changes nothing:

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Reset
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-10-29 19:27 UTC (23):
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                             success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Reset user password                     success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update StsRefreshTokenValidFrom Timestamp success  by Global Admin (IAM admin)
    2026-11-02 19:13  Update user                             success  by Global Admin (IAM admin)
    2026-11-04 17:25  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:25  Update user                             success  by Microsoft password reset service
    2026-11-04 17:25  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:25  Reset user password                     success  by Microsoft password reset service
    2026-11-04 17:25  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:25  Reset password (by admin)               success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)          failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:27  Change user password                    success  by Microsoft password reset service
    2026-11-04 17:27  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:27  Change password (self-service)          success  by sarah.chen@<tenant>
    2026-11-04 17:30  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:30  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                             success  by Azure Credential Configuration Endpoint Service
    2026-11-04 17:34  User registered security info           success  by sarah.chen@<tenant>

DRY RUN  Lost-device reset for Sarah Chen
   1. revoke all sign-in sessions (refresh tokens on every device, including the lost phone)
   2. remove Microsoft Authenticator device: iPhone 12 Pro
      (kept: password, and the password-reset email)

DRY RUN - nothing was changed. Verify the caller's identity, then run with -Apply -VerifiedBy "...".
```

Figure 2. Dry run: 1) revoke all sign-in sessions, 2) remove the Authenticator on the lost iPhone; password and reset e-mail kept.

5. Step 4: the reset

Run with `-Apply -VerifiedBy "Call-back to the phone number HR has on file"`:

Time	Step	Result
20:27:49 UTC	revoke all sign-in sessions	ok
20:27:52 UTC	remove Microsoft Authenticator device: iPhone 12 Pro	ok

```
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-10-29 19:27 UTC (23):
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                             success  by Global Admin (IAM admin)
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    2026-10-29 20:39  Reset user password                     success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update StsRefreshTokenValidFrom Timestamp success  by Global Admin (IAM admin)
    2026-11-02 19:13  Update user                             success  by Global Admin (IAM admin)
    2026-11-04 17:25  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:25  Update user                             success  by Microsoft password reset service
    2026-11-04 17:25  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:25  Reset user password                     success  by Microsoft password reset service
    2026-11-04 17:25  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:25  Reset password (by admin)               success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)          failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:27  Change user password                    success  by Microsoft password reset service
    2026-11-04 17:27  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:27  Change password (self-service)          success  by sarah.chen@<tenant>
    2026-11-04 17:30  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:30  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                             success  by Azure Credential Configuration Endpoint Service
    2026-11-04 17:34  User registered security info           success  by sarah.chen@<tenant>

APPLY  Lost-device reset for Sarah Chen
   1. revoke all sign-in sessions (refresh tokens on every device, including the lost phone)
   2. remove Microsoft Authenticator device: iPhone 12 Pro
      (kept: password, and the password-reset email)
  20:27:49  revoke all sign-in sessions                              ok
  20:27:52  remove Microsoft Authenticator device: iPhone 12 Pro     ok

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-11-05 20:26 UTC (0):

RESET INCOMPLETE - see the FAILED lines above.
```

Figure 3. Reset applied: both steps returned ok. The re-check 3 seconds later still listed the method (see below).

> **Finding: directory replication delay.** Both steps returned **ok**, but the script's own re-check 3 seconds later still listed the Authenticator and printed “RESET INCOMPLETE”. Entra ID can show a deleted method for a short time. A Status check two minutes later confirmed the method was gone. The script was improved: it now waits up to 2 minutes for the change to appear, and only says “incomplete” when a step really failed.

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Status
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (1):
    Password
  MFA methods: 0  (the user will be asked to register at next sign-in)
  Audit events since 2026-10-29 19:29 UTC (26):
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                             success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Reset user password                     success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update StsRefreshTokenValidFrom Timestamp success  by Global Admin (IAM admin)
    2026-11-02 19:13  Update user                             success  by Global Admin (IAM admin)
    2026-11-04 17:25  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:25  Update user                             success  by Microsoft password reset service
    2026-11-04 17:25  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:25  Reset user password                     success  by Microsoft password reset service
    2026-11-04 17:25  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:25  Reset password (by admin)               success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)          failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:27  Change user password                    success  by Microsoft password reset service
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    2026-11-04 17:34  Update user                             success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                             success  by Azure Credential Configuration Endpoint Service
    2026-11-04 17:34  User registered security info           success  by sarah.chen@<tenant>
    2026-11-05 20:27  Update StsRefreshTokenValidFrom Timestamp success  by helpdesk.admin@<tenant>
    2026-11-05 20:27  Update user                             success  by helpdesk.admin@<tenant>
    2026-11-05 20:27  Update user                             success  by Azure Credential Configuration Endpoint Service
```

Figure 4. Status after the reset: Registered methods (1): Password, MFA methods: 0. New audit events at 20:27 by helpdesk.admin: refresh tokens revoked (StsRefreshTokenValidFrom) and the user updated.

6. What the reset did and did not change

Changed	Not changed
All sign-in sessions revoked on every device, including the lost phone	Password (Sarah did not lose it)
Microsoft Authenticator on the lost iPhone removed	Password-reset e-mail
Sarah must register a new MFA method at her next sign-in	Groups, app access, account enabled

7. Step 5: Sarah registers her new phone

Sarah signed in with her password only (no MFA method was left), was asked to register, and added Microsoft Authenticator on the new phone herself. The help desk never handled her credentials.

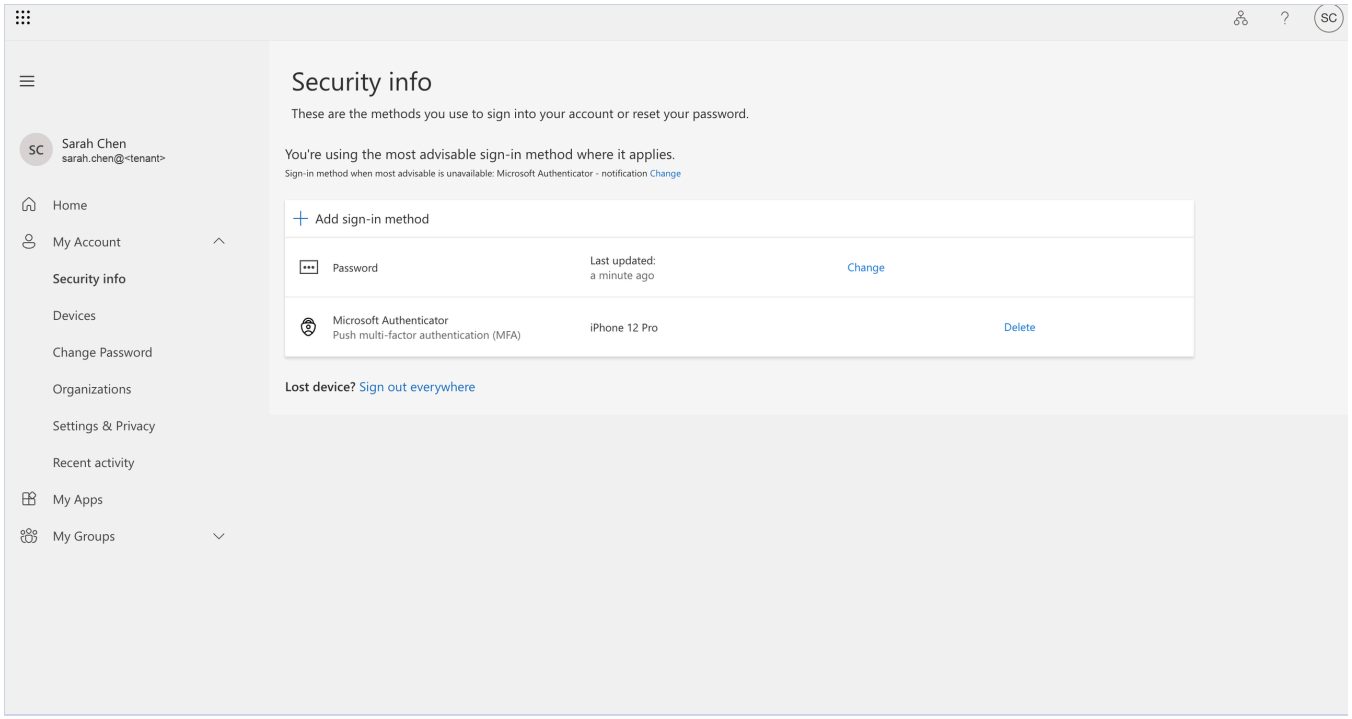


Figure 5. Sarah's Security info page after re-registration: Password and Microsoft Authenticator (push MFA).

> **Lab note.** During the lab, Sarah's password was reset by the IAM admin (3:49 and 3:51 PM) because the test password was not at hand, then changed by Sarah. In a real lost-phone case the password is not touched; it is not part of the MFA reset.

8. Step 6: Verify — 6/6 checks passed

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Verify
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-11-05 20:26 UTC (26):
    2026-11-05 20:27  Update StsRefreshTokenValidFrom Timestamp success  by helpdesk.admin@<tenant>
    2026-11-05 20:27  Update user                                success  by helpdesk.admin@<tenant>
    2026-11-05 20:27  Update user                                success  by Azure Credential Configuration Endpoint Service
    2026-11-05 20:27  Update user                                success  by Azure Credential Configuration Endpoint Service
    2026-11-05 20:27  Admin deleted security info                success  by helpdesk.admin@<tenant>
    2026-11-05 20:49  Update PasswordProfile                     success  by Microsoft password reset service
    2026-11-05 20:49  Update user                                success  by Microsoft password reset service
    2026-11-05 20:49  Update PasswordProfile                     success  by Microsoft password reset service
    2026-11-05 20:49  Reset user password                        success  by Microsoft password reset service
    2026-11-05 20:49  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-05 20:49  Reset password (by admin)                  success  by sabithaskaranavar@<tenant>
    2026-11-05 20:51  Update PasswordProfile                     success  by Microsoft password reset service
    2026-11-05 20:51  Update user                                success  by Microsoft password reset service
    2026-11-05 20:51  Update PasswordProfile                     success  by Microsoft password reset service
    2026-11-05 20:51  Reset user password                        success  by Microsoft password reset service
    2026-11-05 20:51  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
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    2026-11-05 20:51  Update PasswordProfile                     success  by Microsoft password reset service
    2026-11-05 20:51  Change user password                       success  by Microsoft password reset service
    2026-11-05 20:51  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-05 20:51  Change password (self-service)             success  by sarah.chen@<tenant>
    2026-11-05 20:52  Update user                                success  by Azure MFA StrongAuthenticationService
    2026-11-05 20:52  Update user                                success  by Azure MFA StrongAuthenticationService
    2026-11-05 20:52  Update user                                success  by Azure Credential Configuration Endpoint Service
    2026-11-05 20:52  User registered security info              success  by sarah.chen@<tenant>
    2026-11-05 20:52  Update user                                success  by Azure Credential Configuration Endpoint Service

  PASS  identity verified before the reset
  PASS  old MFA methods removed at reset
  PASS  new MFA method registered after the reset
  PASS  audit: admin deleted security info
  PASS  audit: user registered security info
  PASS  account still enabled

6/6 checks passed
```

Figure 6. Verify run at 3:54 PM: all 6 checks passed, including “Admin deleted security info” by helpdesk.admin and “User registered security info” by Sarah.

9. Step 7: audit log in the Entra admin center

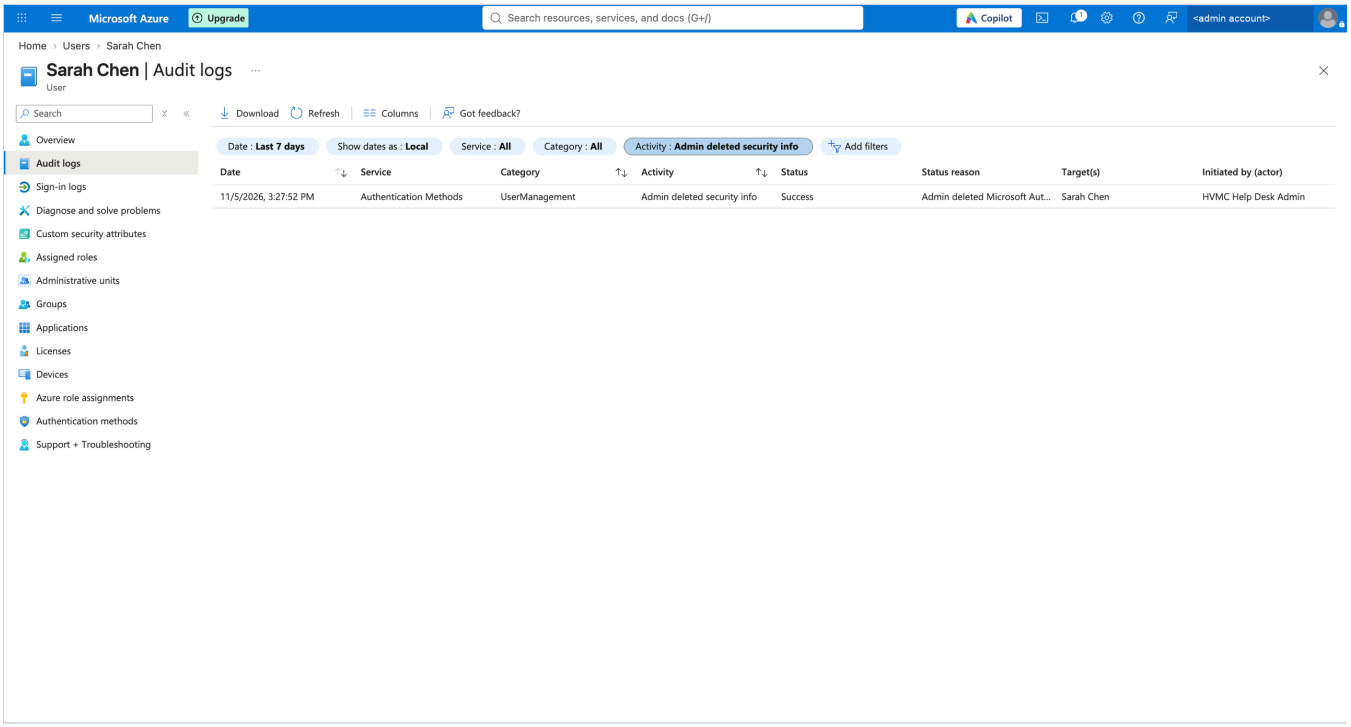


Figure 7. "Admin deleted security info" (Microsoft Authenticator) at 3:27 PM, initiated by HVMC Help Desk Admin, not the Global Admin.

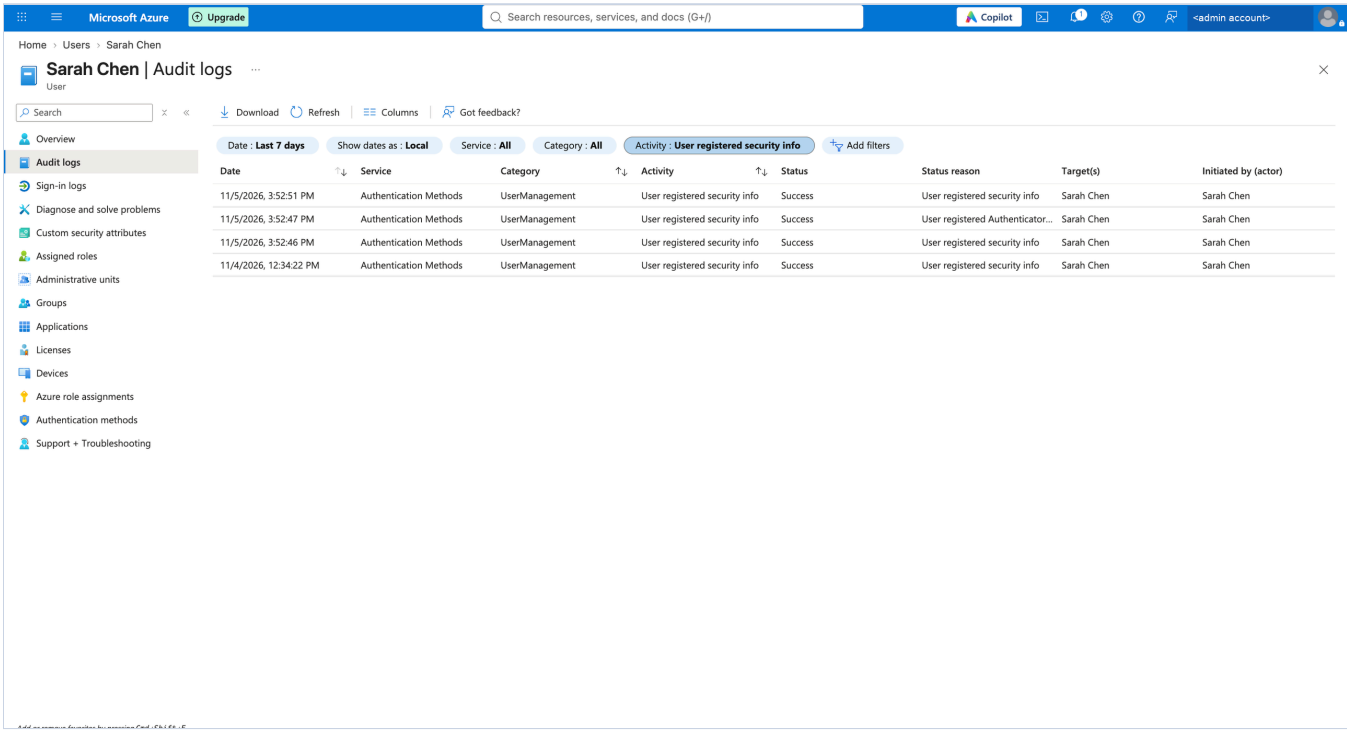


Figure 8. "User registered security info" by Sarah Chen: the old phone on 4 Nov and the new phone on 5 Nov at 3:52 PM.

10. Result

Item	Result
Ticket	IAM-0004 opened 3:25 PM, closed 3:54 PM (29 minutes), see Ticket_IAM-0004_MFA_Support.pdf
Lost phone	Signed out everywhere and its Authenticator removed
User	Back in with MFA on the new phone; password, groups and apps unchanged by the MFA reset
Who did it	helpdesk.admin (Authentication Administrator + Reports Reader), with the caller's identity checked first
Evidence	Script evidence JSON, 9 screenshots, Entra audit log (Admin deleted / User registered security info)